

TARUN JOSEPH

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SUMMARY

Machine Learning Engineer and Graduate Teaching Assistant with four years of industry experience implementing AI-powered solutions. Currently supporting first-year and graduate-level engineering courses at the University of Toronto while preparing for an upcoming internship at the Vector Institute. Proven record of managing technical deliverables, leading large-scale training bootcamps, and handling departmental leadership roles within the university community.

TECHNICAL SKILLS

Programming: Python, R, SQL, Java, C++

AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, LangChain

Data Tools: Pandas, NumPy, Tableau, MySQL, PostgreSQL, pgAdmin, SciPy, Statsmodels

Cloud Platforms: AWS, Azure

Web Development: Flask, HTML, CSS

Other Tools & Skills: JIRA, Git, Power BI, BeautifulSoup, Selenium, Agentic Frameworks

EDUCATION

➤ **MEng, Industrial Engineering | University of Toronto, Canada** *September 2025 – Present*

Emphasis: Data Analytics & Machine Learning, Entrepreneurship and Management (ELITE)

Relevant Courses: MIE1626H: Data Science Methods and Statistical Learning (A+), MIE1517: Deep Learning (A), APS1070: Foundations of Data Analytics and Machine Learning (A), ECE1786H: Applications of NLP (upcoming)

➤ **BTech, Industrial Engineering | APJ Abdul Kalam Technological University | College of Engineering Trivandrum (CET), India** *August 2017 – June 2021*

Relevant Courses: Operations Research, Data Analysis using R and Python

TEACHING EXPERIENCE

➤ **Teaching Assistant | APS106: Fundamentals of Computer Programming | University of Toronto** *(January 2026 – April 2026)*

- Supervising in-person labs and tutorials for 500+ students, focusing on the practical implementation of algorithms in Python.
- Guiding students through the transition from basic syntax to complex structure-based and object-oriented programming.

➤ **Teaching Assistant | MIE1628: Cloud Based Big Data Analysis | University of Toronto** *(January 2026 – April 2026)*

- Assisting graduate students with cloud-based data processing, technical troubleshooting, and big data project implementations.
- Provided feedback on machine learning model implementations and data analysis pipelines as part of graduate-level assignments.
- Managing grading and providing technical feedback on midterms and assignments to ensure student competency in core data engineering and cloud concepts.

➤ **Teaching Assistant | MIE 429: Machine Intelligence AI Capstone Design | University of Toronto** *(September 2025 – December 2025)*

- Mentored 45 final-year students through AI capstone projects, providing feedback on algorithm design and Python implementation.

➤ **Subject Matter Expert (SME) | Tata Consultancy Services (TCS)** *(January 2023 – May 2025)*

Delivered training as a Subject Matter Expert (SME) for the following organization-wide upskilling programs:

- **AI Engineer Program:** AI Ethics, NLP, Computer Vision
- **Generative AI Bootcamp:** Prompt Engineering, Retrieval Augmented Generation (RAG), LLM Application Development, AI Agents, Knowledge Graphs
- **Data Scientist Program:** Supervised and Unsupervised Learning in Machine Learning
- **Freshers Training Bootcamp:** HTML, CSS, JS, Python, Flask

➤ **TATA ProEngage NGO | Skill Upgradation for Rural Youth Empowerment (SURE)**

❖ **Lead Trainer & Mentor**

March 2025 – August 2025

- Delivered 3-hour online sessions weekly as part of a 6-month technical bootcamp for 20 final-year CS/IT students, focusing on Python programming, AI, and machine learning fundamentals.
- Mentored students individually in developing industry-relevant projects to strengthen internship prospects and support transitions into technical careers.

➤ **Make A Difference (MAD)**

❖ **Academic Support Volunteer**

October 2019 – June 2021

- Provided academic tutoring in fundamental Math and English to 10+ underprivileged children (ages 10–15) in shelter homes, through both in-person and online sessions.
- Co-developed personalized learning plans and interactive teaching methods to improve academic engagement, confidence, and foundational understanding.
- Collaborated with a team of volunteers to track student progress and adapt support strategies based on individual learning needs.

PROFESSIONAL EXPERIENCE

➤ **Applied AI Intern | Vector Institute, Toronto**

May 2026- Present

- Serving as a Design and Delivery (D&D) Applied AI Intern, bridging the gap between complex AI research and practical industry implementation.
- Acting as a technical translator to communicate project requirements and facilitate smooth transitions for non-technical industry sponsors.

➤ **Machine Learning Engineer | Tata Consultancy Services (TCS)**

January 2023 – May 2025

- Deployed the following ML solutions for TCS's Edtech platform, enhancing learning experiences for over 600,000 associates:
 - GPT-3.5 Large Language Model (LLM) powered Retrieval-Augmented Generation (RAG) chatbot for coding assistance and Q&A.
 - Llama 3 LLM-powered AI Agent for ticket automation.
 - Web Scraping/OCR tool for efficient data mining and extraction.
- Organized weekly and monthly AI/ML hackathons and coding competitions, driving active learning and significantly boosting program registrations.
- Designed AI/ML coding assessments for over 10,000 associates per quarter, supporting career advancement through promotions.
- Automated redundant manual tasks using AI tools, improving developer and learner productivity across teams.
- Led deep-skilling bootcamps on Generative AI, LLMs, AI/ML, and Python, increasing platform engagement metrics (Learning-Hours) by 13% in FY24.

➤ **Data Engineer | Tata Consultancy Services (TCS)**

May – December 2022

- Owned and managed data reporting operations for TCS Talent Development's (TD) senior engineer program, ensuring seamless data flow from the database to the front-end portal.

- Developed tables and pipelines using PL/SQL to optimize data migration, storage, and warehousing operations.
- Delivered 30+ user stories and revamped 5+ employee data reports, enabling frictionless data extraction and business analysis operations.
- Executed ETL activities and conducted database monitoring/maintenance using Informatica and pgAdmin, improving resiliency and minimizing downtime.
- Collaborated with business analysts to gather and address requirements effectively, achieving a customer satisfaction rating of 8 in FY23 Q3.

➤ **Data Analyst | Tata Consultancy Services (TCS)**

January – April 2022

- Developed a unified SQL database and Tableau dashboard for program data analysis and automated recommendation mailers using Python, reducing manual effort by around 50%.
- Analyzed platform data to derive insights for talent development programs, addressing the declining completion-enrollment ratio and driving an 11% increase in engagement metrics (Learning Hours) in FY22.
- Received 'On the Spot' award for providing innovative insights and actionable feedback to senior management, derived from past enrollment data and candidate interviews, resulting in improved program registration numbers.

RESEARCH EXPERIENCE/PROJECTS

➤ **University of Toronto | American Sign Language (ASL) Real-Time Translator (2026)**

- Leading a four-person team to design a Deep Learning-based ASL translator intended to facilitate real-time conversations for the deaf and hard-of-hearing community.
- Implementing computer vision models to accurately recognize and translate hand gestures into text/speech.

➤ **Tata Consultancy Services (TCS)**

❖ **Deepfake Classification and Captioning using Transformers**

- Developed a global hackathon challenge leveraging CNN and Transformer-based Image Classifier and Captioning systems to distinguish between AI and human-generated images while generating precise descriptions.
- Organized the hackathon as part of TCS's yearly learning festival, attracting nearly 5,000 participants and ensuring comprehensive visibility across the organization.

❖ **Ticket Automation using Generative AI Agent**

- Presented Proof of Concept (POC) and implemented a Minimum Viable Product (MVP) for support ticket automation using open-source LLM models and agentic frameworks.
- Achieved significant efficiency gains by eliminating redundant developer efforts, saving over 5 hours per week.

➤ **College of Engineering Trivandrum (CET)**

❖ **Analysis of Telecom Customer Churn during Pandemic Period**

- Analyzed and predicted the impact of the COVID-19 pandemic on a regional telecom operator using machine learning algorithms, achieving an 83% accuracy in churn prediction.
- Provided actionable insights that enabled the company to implement two new customer retention schemes, resulting in a 35% increase in quarterly revenue by retaining high-value customers and onboarding new ones.

❖ **Deep Learning in Medical Imaging**

- Researched and published a seminar report on the applications of neural networks in MRI image processing and healthcare data analytics.
- Identified key challenges in data availability and proposed innovative strategies to address them, enabling more efficient utilization of healthcare datasets.

LEADERSHIP & COMMUNITY INVOLVEMENT

- **VP of Professional Development | MIE Graduate Student Association (AMIGAS) (2025 – Present)**
 - Managing professional development initiatives and coordinating industry-focused events for the graduate student body.
- **Departmental Steward | CUPE 3902 Unit 1 (2025 – Present):**
 - Serving as a liaison between the union and the department to ensure adherence to professional standards and the collective agreement.

CERTIFICATIONS/ACHIEVEMENTS

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| • IIT Kharagpur AI4ICPS Certificate Program | <i>July - October 2024</i> |
| • IIMB Generative AI and Large Language Models | <i>July - October 2024</i> |
| • AWS Certified AI Practitioner | <i>October 2024</i> |
| • Microsoft Certified Azure Data Scientist Associate | <i>May 2024</i> |
| • PCAP-Certified Associate in Python Programming | <i>May 2023</i> |
| • TCS Digital (Hi-Talent) Cadre | <i>October 2022</i> |
| • TCS On the Spot Award | <i>March 2022</i> |

STANDARDIZED TEST SCORES

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| ➤ IELTS | <i>22nd September 2024</i> |
| • Overall Band Score: 8.0 | |
| • Listening: 9.0, Reading: 8.5, Writing: 7.0, Speaking: 7.0 | |
| • CEFR Level: C1 | |
| ➤ GRE | <i>9th November 2024</i> |
| • Verbal Reasoning: 150 | |
| • Quantitative Reasoning: 161 | |
| • Analytical Writing: 3.5 | |
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